

Replacement-Based Privacy Protection Scheme in RAG Scenarios

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Abstract

Large Language Models (LLMs) have substantially enhanced work efficiency due to their advanced semantic comprehension and training on extensive datasets. However, outdated training corpora and the limited availability of domain-specific data in many general-purpose models often result in suboptimal performance in specialized fields. Retrieval-Augmented Generation (RAG) has emerged as an effective technique for improving model output quality by integrating external knowledge sources, enabling models to retrieve relevant information prior to inference. Nevertheless, leveraging internal knowledge bases introduces significant privacy concerns, as these repositories often contain sensitive information, increasing the risk of unintended data exposure. To address this issue, we propose EntitySafe, a privacy-preserving synthetic data generation framework based on controlled text replacement. EntitySafe comprises two primary modules: numerical perturbation and entity obfuscation. The numerical perturbation module employs LLMs in conjunction with differential privacy mechanisms to synthesize numerical data, ensuring privacy guarantees while maintaining utility. The entity obfuscation module leverages Natural Language Processing (NLP) techniques to systematically replace sensitive entities in retrieved contexts with contextually appropriate substitutes. Experimental evaluations demonstrate that EntitySafe-generated synthetic data preserves the effectiveness of RAG while significantly mitigating privacy risks.

Keywords: RAG, Privacy Protection, Synthetic Data Generation, EntitySafe

1. Introduction

In 2017, Google (Vaswani, 2017) introduced the Transformer architecture, a novel design entirely based on the self-attention mechanism. This architecture superseded traditional Recurrent Neural Network (RNN) structures, enabling parallel processing of sequential data and significantly improving training efficiency. The Transformer's ability to handle long-range dependencies enhanced model performance and paved the way for subsequent pre-trained models. Its success has cemented the Transformer as the foundational framework for virtually all mainstream LLMs. Following the Transformer's success, pre-training and fine-tuning strategies rapidly became prevalent in the field of Natural Language Processing (NLP). Models such as BERT (Devlin et al., 2018) (Bidirectional Encoder Representations from Transformers) and GPT (Radford et al., 2018) (Generative Pre-trained Transformer) have achieved unprecedented results through pre-training on massive text datasets and subsequent fine-tuning for specific tasks. This approach not only improves a model's generalization capabilities but also substantially reduces the need for labeled data (Radford et al., 2019).

It has been observed (Petroni, 2019) that models can store rich relational knowledge from their training data, much like a knowledge base. However, since the model's training relies on a fundamentally stable and large dataset, these data collectively form the "brain" of the model. This makes it difficult to expand or modify the model once it is trained. As a result, when faced with questions that go beyond the model's training data, it is prone to generating incorrect or misleading answers. In this way, the training dataset becomes a boundary for the model's

capabilities (Miller, 2016).

A hybrid model that combines parametric and non-parametric memory can alleviate this issue to some extent. In such a model, the externally constructed knowledge base from non-parametric memory serves as the model's "external brain," providing additional knowledge support. This concept is closely related to the modern approach of Retrieval-Augmented Generation (RAG), introduced by Patrick Lewis et al. in 2020 (Lewis, 2020).

RAG represents a hybrid approach that integrates information retrieval (IR) with language generation models. It aims to overcome the limitations of purely Transformer-based language models in handling long-tail factual knowledge and the need for a large number of parameters to retain comprehensive world knowledge. By integrating a retrieval component with a generation component, RAG enables dynamic access to external information sources during inference, thereby enhancing the model's knowledge and adaptability. A typical RAG system comprises a retrieval component and a generation component. The retrieval component searches for relevant content within an external knowledge base based on user input using specific algorithms, and then integrates the retrieved results with the user's input. The generation component then produces the final output.

However, recent studies (Zhang, 2021) have shown that current RAG systems are highly vulnerable to sensitivity attacks. These systems can be easily manipulated to retrieve and output sensitive information by crafting prompts that include both "information" and "commands." This risk of information leakage significantly restricts the practical application of RAG systems. Consequently, substantial research efforts have focused on en-

ensuring that these models can provide high-quality responses while adhering to strict privacy requirements (Al Aziz et al., 2021; Hathurusinghe et al., 2021; Cai et al., 2021; Zeng et al., 2024c).

One significant contributor to this risk is the inherent identifiability of entities, particularly named entities. These entities serve as identifiers for real-world individuals, organizations, and locations, making them highly recognizable (Nadeau and Sekine, 2007). Legal frameworks like GDPR and HIPAA recognize this link by defining Personally Identifiable Information (PII) to include elements often expressed as named entities, such as names, addresses, and contact details. Furthermore, studies on re-identification attacks (Narayanan and Shmatikov, 2008) demonstrate how seemingly innocuous entities can be combined to reveal the identities of individuals even in anonymized datasets, highlighting the fundamental privacy risk associated with their unique and distinguishing nature.

In 2024, Zeng et al. proposed SAGE (Zeng et al., 2024c) represents a significant step forward in addressing privacy concerns in RAG. SAGE is a two-stage synthetic data generation paradigm for RAG. While SAGE demonstrates promising results in mitigating privacy risks by generating synthetic data, it relies on closed-source LLMs in its initial stages, raising new privacy concerns. Furthermore, its reliance on direct data transmission to these models and certain limitations in controlling data generation warrant further investigation.

To mitigate the fundamental privacy risk, overcome the limitations of the SAGE framework, and meet the demand for privacy-preserving RAG frameworks, we propose EntitySafe—a synthetic data generation framework built on two key techniques: numerical perturbation and entity replacement.

We implement a probabilistic replacement strategy for sensitive entities and data. Each entity in the text is independently subjected to a replacement decision, with a probability p of being replaced by a synthetic alternative. The value of p is configurable and can be adjusted based on the desired level of privacy protection. This probabilistic approach implies that, in a sufficiently large sample of text, we expect approximately $1-p$ of the original entities to remain unchanged, contributing to the overall utility of the retrieval content. Furthermore, we can protect against attacks that leverage combined quasi-identifiers by implementing probabilistic replacement. A quasi-identifier is a set of attributes that, when combined, can potentially identify an individual with a high degree of certainty. For example, an attacker might attempt to identify a patient by combining their general residential area, the nature of their medical condition, and the approximate time of a medical visit. Let p_i be the probability that the i -th entity in the combined identifier is replaced with a synthetic value. The probability that the entire combined identifier remains authentic is then given by $\prod_{i=1}^n (1 - p_i)$, where n is the number of entities in the combined identifier.

Crucially, since each attribute/entity is independently subjected to a probabilistic replacement, the attacker faces an inference chain where the confidence in each individual attribute/entity is inherently limited. As the length of the inference chain, n , increases, the overall confidence in any derived conclusion diminishes exponentially, trending towards com-

plete untrustworthiness. This can be formally represented as $\lim_{n \rightarrow \infty} \prod_{i=1}^n (1 - p_i) = 0$, given that $0 < p_i \leq 1$ for at least one i .

By probabilistically replacing or perturbing these quasi-identifiers, we prevent attackers from accurately reconstructing the sensitive information associated with an individual.

This paper makes the following contributions:

- We propose a novel and adaptable data synthesis method, EntitySafe, tailored for protecting privacy in RAG scenarios. EntitySafe offers a balance between privacy and utility by leveraging probabilistic replacement and LLMs, ensuring RAG systems can operate on sensitive data without exposing private information. Unlike existing methods, EntitySafe offers a flexible framework, through parameter adjustment, to adapt to variable security needs for both public, and private domains.
- We introduce a numerical perturbation method based on differential privacy, designed to enhance the reasonableness and credibility of numerical data within synthetic RAG contexts.

The core contribution of this paper is the EntitySafe framework, which provides a novel approach to generating high-quality, privacy-preserving synthetic data for RAG systems. This framework combines numerical perturbation and entity replacement techniques to achieve a strong balance between privacy and utility.

This paper is organized as follows: In Section 2, we introduce the related work, highlighting the key advancements in the field. Section 3 details the specific methodology employed in our study, providing a comprehensive overview of the techniques and approaches used. In Section 4, we present the experiments conducted and the results obtained, demonstrating the effectiveness of our methodology. Finally, Section 5 concludes the paper, summarizing our findings.

2. Related work

Extensive research has investigated privacy data leakage in LLMs and RAG systems. Carlini et al. (Carlini et al., 2023) examined the memorization capabilities of LLMs concerning their training data, identifying three factors influencing the degree of memorization. They demonstrated that LLMs can memorize substantial amounts of training data, raising privacy and fairness concerns. Similarly, other studies (Kandpal et al., 2022) have revealed the susceptibility of LLMs to privacy attacks. Zeng et al. (Zeng et al., 2024a) focused on privacy leakage in fine-tuned models, examining their memory behavior across tasks and highlighting vulnerabilities in areas like dialogue. Further, Zeng et al. (Zeng et al., 2024b) introduced targeted and untargeted attack methods to assess retrieval data leakage risks within RAG systems, empirically validating their effectiveness and highlighting the vulnerability of current RAG systems. Consequently, diverse methods for safeguarding model data have been proposed, with data anonymization

involving sensitive information replacement as a prevalent approach (Miranda et al., 2024).

Altieri et al. (Zuo et al., 2021) reviewed data anonymization techniques in digital healthcare, mentioning strategies like data perturbation, data generalization, data masking, and differential privacy. However, these techniques primarily handle structured tabular data, rendering them inadequate for processing the unstructured text common in RAG scenarios.

To address this, Zhou et al. (Zhou et al., 2022) introduced DataSifterText, a method for generating partially synthetic clinical text data that balances utility and privacy. DataSifterText is specifically designed for clinical text and utilizes BERT models to predict and replace content, focusing on obfuscating the location of sensitive information. In contrast, EntitySafe is intended for broader RAG applications, employing numerical perturbation with differential privacy for numerical data and NLP-based entity replacement for textual data, allowing for nuanced control over privacy and utility across diverse data types. DataSifterText achieves its goal by obfuscating the location of sensitive information and utilizing BERT models to predict and replace portions of the content, thus generating synthetic text with a similar distribution to the original.

Vats et al. (Vats et al., 2024) explore the potential for LLMs to recover masked sensitive information, underscoring the need for robust masking and substitution strategies. While Vats et al. analyze potential vulnerabilities, EntitySafe employs a complementary proactive approach, generating synthetic data to mitigate privacy risks before data exposure and ensuring utility for both retrieval and generation within RAG pipelines. Romanov et al. (Romanov et al., 2019) presented a technique for reducing the accuracy of author attribution systems, protecting text authors’ identification. While Romanov et al. focus on anonymizing the style of the text, EntitySafe addresses the content to prevent privacy leakage within a RAG system. Romanov’s work doesn’t address the risks associated with private data disclosure within the text, a core concern for EntitySafe.

To further address these challenges, Zeng et al. (Zeng et al., 2024c) introduced SAGE, a two-stage framework for generating synthetic data to protect privacy in RAG systems. SAGE leverages attribute-based extraction and generation in the first stage, using LLMs to identify key attributes and generate synthetic data. In the second stage, SAGE employs an agent-based iterative refinement process. While SAGE demonstrates promising results, it also requires directly passing sensitive data to closed-source LLMs under unprocessed conditions to obtain key attributes, raising privacy concerns during attribute extraction and synthetic data generation. The refinement process also depends heavily on the capabilities of the LLMs used and may not fully eliminate privacy risks. Further, the synthetic data generated may not fully capture the nuanced semantic information, potentially leading to a utility loss. In contrast, EntitySafe combines numerical perturbation with differential privacy to protect numerical attributes and leverages NLP techniques for entity replacement, avoiding the direct exposure of privacy data and minimizing the risk of exposing sensitive data.

3. Method

To mitigate potential privacy concerns within RAG systems, and drawing inspiration from the SAGE framework Zeng et al. (2024c), we propose EntitySafe. Unlike SAGE, EntitySafe uses a numerical perturbation method combined with differential privacy to protect numerical information that may appear in various forms within natural language text and leverages NLP for entity replacement. This avoids reliance on closed-source LLMs for data transformation and eliminates the complexities associated with dual-agent rewriting mechanisms. The core principle of EntitySafe lies in the synthetic processing of retrieval data, generating alternative data that preserves privacy while maintaining utility. The framework comprises two primary components: a numerical perturbation component and an entity replacement component. The numerical perturbation component generates a context containing perturbed numerical values. This modified context is then passed to the entity replacement component, which either rephrases the perturbed numerical values or performs entity replacement. To further mitigate privacy risks associated with utilizing LLMs, our method incorporates multiple safeguards. First, during the entity replacement stage, we strictly limit the LLM’s role to generating candidate replacements. The LLM only receives a large set of unordered entities and generates candidate replacements for them, avoiding direct contact with raw sensitive data. Second, in the numerical perturbation stage, the LLM interacts only with the masked numerical values and the context section containing the mask, which is limited in length, minimizing unnecessary information release. Moreover, we apply differential privacy with carefully chosen parameters (σ , δ) to protect the privacy of individual numerical values. We also select reputable LLM APIs and rigorously examine their privacy policies to ensure responsible data handling. Finally, we are actively exploring the possibility of using locally deployed LLMs, with the goal of achieving complete control over data processing. The entire Protection framework of EntitySafe as shown in Figure 1.

3.1. Part One: Numerical perturbation

This section details our approach to perturbing numerical components within the knowledge base. These numerical values, representing attributes like time, addresses, or blood pressure, are perturbed to defend against privacy attacks that could arise if the RAG system directly outputted sensitive numerical values. To ensure the RAG system accesses only protected data, our method permanently modifies the original text within the knowledge base, guaranteeing that it retrieves the protected version and preventing direct exposure of sensitive numerical information. The unstructured nature of surrounding text and the human-typed content within retrieval datasets can compromise data accuracy and reliability due to potential typing biases, leading to chaotic numerical representations. The complexity of natural language adds to this challenge, making it difficult to ascertain which entity a number modifies in context. Existing NLP techniques often struggle to accurately identify the entity a numerical value pertains to. Therefore, we decided to leverage the strong contextual awareness of large language models

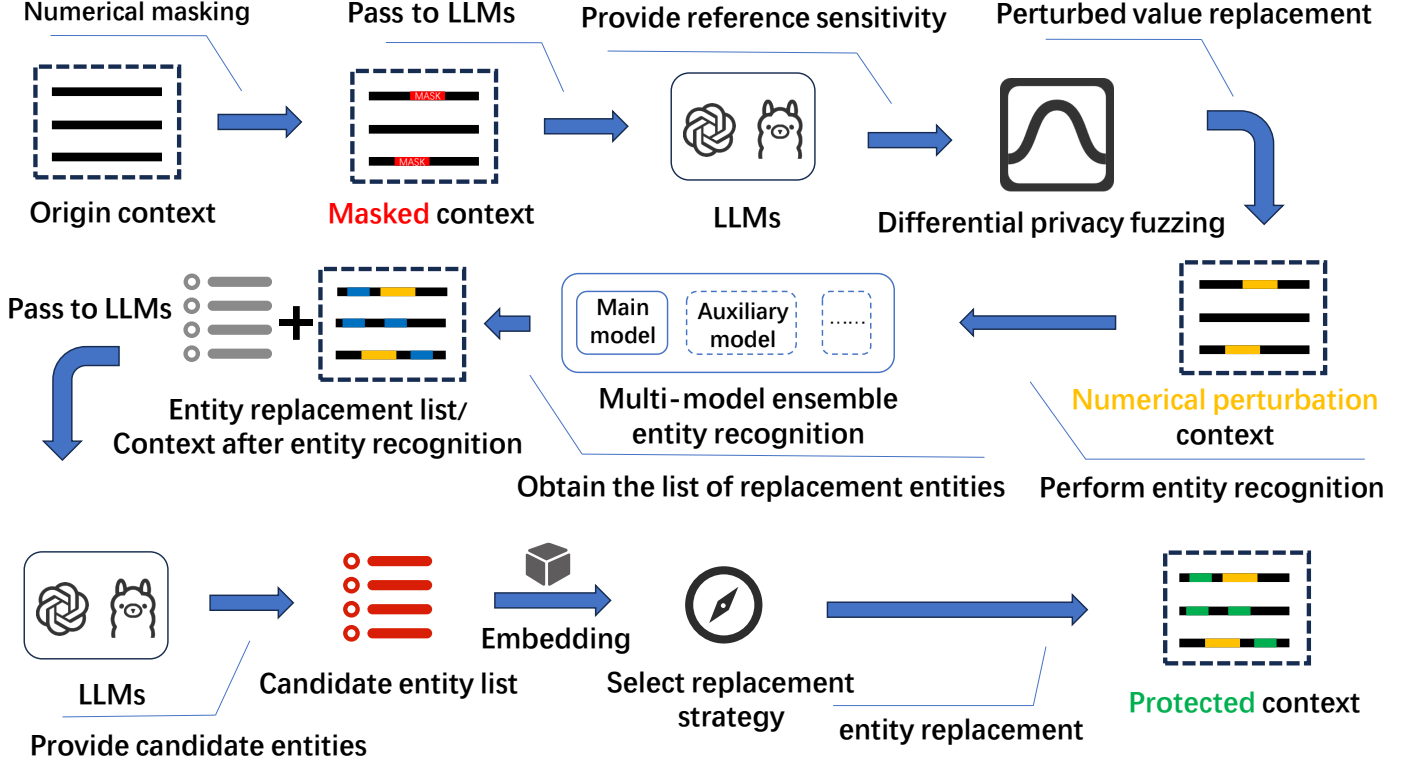


Figure 1: Protection framework of EntitySafe. Here, *Entity replacement list* refers to the list of entities identified in the text that need to be replaced. *Candidate entity list* refers to a two-dimensional list composed of candidate replacement lists generated by the model for each entity that needs to be replaced.

to determine the entities modified by masked numerical values within complex semantic contexts, thereby establishing correspondences between numerical values and their associated entities. Furthermore, the acceptable perturbation range is limited due to the variability of modifying entities. For instance, perturbations of age, blood pressure, and blood glucose levels must fall within different ranges. The model can provide reference ranges or users can leverage their domain knowledge to calculate these ranges. To prevent direct exposure of private content to a closed-source large model, we first mask the content. SAGE aims to generate synthetic data free of PII, relying on the LLM’s ability to avoid generating sensitive information. This approach may not guarantee sufficient privacy, as LLMs can inadvertently leak PII or possess imperfect safety alignments. Conversely, our method proactively masks PII before submitting text to the LLM, preventing the LLM from ever accessing sensitive information. SAGE’s agent-based refinement process only checks for privacy violations after data generation, potentially exposing the LLM to PII. Our method, however, provides privacy protection before data generation, eliminating this risk.

Based on the above considerations, we propose a four-stage numerical perturbation strategy:

1. **Masking:** Mask the numbers in the original contextual text.
2. **Model Input:** Submit the masked text to a large language model (LLM). The LLM is then used to: (1) establish the correspondence between masked numerical values and

their modifying entities, (2) provide a reference sensitivity score for each numerical value, reflecting its potential privacy risk, (3) determine whether sign changes are permissible during perturbation, and (4) determine whether the numerical value is suitable for modification. However, if the organization already possesses a substantial amount of data for a particular entity (e.g., a hospital holding a large dataset of patient blood pressure readings), the reference sensitivity score from the LLM can be bypassed. Instead, the organization can leverage the LLM’s provided numerical value-entity relationship to calculate the sensitivity based on aggregated data and statistical information derived from its own data, adhering to strict privacy protocols. For entities where the organization lacks sufficient data, the LLM’s reference sensitivity score is utilized. The LLM provides this reference sensitivity score based on the masked text, leveraging contextual information, domain knowledge, and pre-defined privacy policies without directly accessing the original numerical values. Even in the worst-case scenario where the organization lacks sufficient data or the LLM assessment is inaccurate, our method still provides a baseline level of privacy protection through the differential privacy mechanism. The LLM considers the nature of the numerical value and constrains the range of the perturbation accordingly to maintain data validity.

3. **Differential Privacy Calculation:** Combine the obtained pre-calculated sensitivity with other parameters to calculate specific differential privacy variance. The pre-

calculated sensitivity, denoted as Δf , is then used to determine the noise parameter σ in the differential privacy mechanism (Equation 2), ensuring that the added noise effectively protects privacy while preserving data utility. Applying Gaussian noise generated from differential privacy variance to the original numbers.

4. Replacement: Replace the masks with the corresponding perturbed numbers, applying the previously determined sign and range constraints.

The entire numerical perturbation process as shown in Figure 2.

Applying Gaussian noise generated from differential privacy replacement to the original numbers. The probabilistic entity replacement strategy, governed by the parameter p , provides an additional layer of defense against sophisticated attacks. Even if an attacker possesses auxiliary information about a particular entity or attempts to leverage differential queries, the inherent uncertainty introduced by the replacement process significantly hinders their ability to construct a reliable inference chain. An attacker faces 2^n possible combinations of original and replaced entities, where n is entities with replacement options. The probability p dictates the likelihood that any given entity has been replaced, thus disrupting any deterministic relationships between entities that the attacker might attempt to exploit. This forces the attacker to consider a vast space of possibilities, as they must account for both the original and replaced values for each entity. Moreover, the replaced entities may introduce misleading information, leading the attacker down incorrect paths and ultimately thwarting their efforts to infer sensitive information. Therefore, by combining differential privacy with probabilistic entity replacement, we create a robust privacy-preserving mechanism that effectively protects against a wide range of attacks.

3.1.1. Differential privacy

Differential privacy provides a framework to quantify the privacy guarantees of algorithms that analyze sensitive data. To achieve (ϵ, δ) -differential privacy, we can add Gaussian noise to the output of a function f . The mechanism is defined as follows:

$$\mathcal{M}(x) = f(x) + \mathcal{N}(0, \sigma^2 I) \quad (1)$$

where $\mathcal{N}(0, \sigma^2 I)$ denotes Gaussian noise with mean 0 and variance σ^2 , and I is the identity matrix.

The choice of σ is crucial and is determined by the sensitivity Δf of the function f and the privacy parameters ϵ and δ :

$$\sigma \geq \frac{\Delta f}{\epsilon} \sqrt{2 \ln(1.25/\delta)} \quad (2)$$

This ensures that the probability of any single individual’s data significantly affecting the output is minimized, thus providing privacy guarantees.

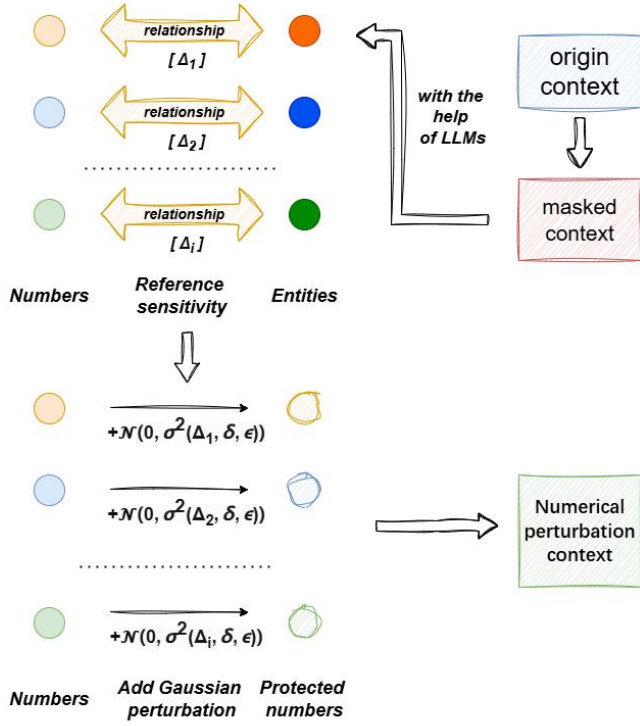
3.2. Part Two: Entity Replacement

Building upon the numerically perturbed context from the first stage, we employ Named Entity Recognition (NER) technology to identify entities. These entities are then submitted to a large model, which randomly selects five perspectives – including randomization, obfuscation, differential privacy, semantic abstraction, fictional replacement, synonym replacement, and domain-specific variation – to generate candidate replacements for each entity. For polysemous entities, the model prioritizes ensuring at least one replacement angle exists for each meaning. For each entity, we attempt to replace the original entity within the numerically perturbed context using each candidate from its corresponding replacement list and calculate the semantic similarity of all replacements. A specific strategy then dictates the final replacement, offering flexibility based on the application and desired privacy level. In our experiments, we selected the second-most similar replacement from the list for each entity. This ensures the resulting text differs from the original while maintaining strong semantic relevance and utility, helping to prevent simple re-identification attacks and preserving the usefulness of the information for the RAG system. To enhance entity recognition effectiveness, we explore employing an ensemble of multiple spaCy models. This involves simultaneously using a main model and a secondary model, while also considering context nouns as replaceable objects. The main model identifies common entities, while the secondary model supplements the main model by recognizing additional entities, particularly within specialized domains such as medical data, where a fine-tuned secondary model can identify specific medical content that a general model might miss.

Finally, we use a hyperparameter called “replacement rate” to control the desired level of privacy. We designed the primary model to specifically identify known and high-risk privacy-sensitive entities, such as names and phone numbers. The leakage of these entities can easily lead to identity exposure and misuse of personal information, thus necessitating enforced 100% replacement. However, the entities and nouns identified by the secondary model may contain semantically important information, but not all of them necessarily involve privacy leakage. Replacing all of these entities may excessively reduce the utility of the data and affect the retrieval performance of the RAG system. Therefore, we adopt a probabilistic replacement strategy, replacing only a proportion of the entities and nouns identified by the secondary model to strike a balance between privacy protection and data utility. This “replacement rate” indicates the proportion of entities and nouns identified by the secondary model that are actually replaced. Users can adjust this rate based on specific privacy requirements. A higher replacement rate offers stronger privacy protection, but may reduce data utility. Therefore, selecting the appropriate replacement rate requires balancing privacy and utility. The flow of the entity replacement part is shown in Figure 3.

4. Experiments

This section details the experiments conducted to evaluate EntitySafe’s effectiveness in terms of RAG performance and



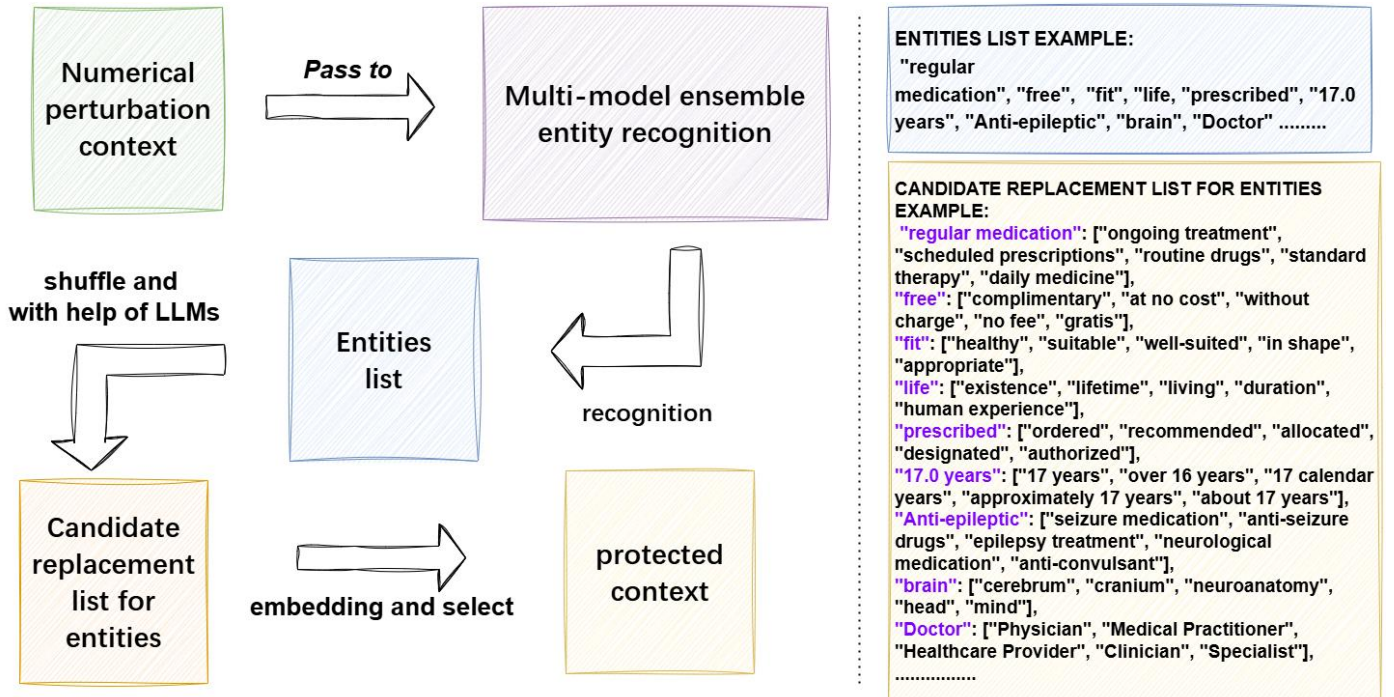
EXAMPLE

Hi Im 38 yo male of about 145 lbs in weight. Just got my blood test results which indicate high triglycerides of 203.58 (reference value 60-165 mg/dl) and VLDL-cholesterol of 40.54 (reference value of 0-40 mg/dl). The rest of the tests came out normal with quite low SGPT of 31.17 (ref value 0-49) and SGOT of 26.21 (ref value 0-46). Should I be alarmed? Thanks!

Hi Im [MASK] yo male of about [MASK] lbs in weight. Just got my blood test results which indicate high triglycerides of [MASK] (reference value [MASK]-[MASK] mg/dl) and VLDL-cholesterol of [MASK] (reference value of [MASK]-[MASK] mg/dl). The rest of the tests came out normal with quite low SGPT of [MASK] (ref value [MASK]-[MASK]) and SGOT of [MASK] (ref value [MASK]-[MASK]). Should I be alarmed? Thanks!

Hi Im 39 yo male of about 139 lbs in weight. Just got my blood test results which indicate high triglycerides of 214.79 (reference value 71-170 mg/dl) and VLDL-cholesterol of 38.68 (reference value of 0-36 mg/dl). The rest of the tests came out normal with quite low SGPT of 31.05 (ref value 1-52) and SGOT of 25.17 (ref value 0-46). Should I be alarmed? Thanks!

Figure 2: In this context, *origin context* represents the text content selected during the RAG retrieval phase to be enhanced. *masked context* refers to the text content after masking the numerical values. *numerical perturbation context* represents the text content after undergoing numerical perturbation.



ENTITIES LIST EXAMPLE:

"regular medication", "free", "fit", "life", "prescribed", "17.0 years", "Anti-epileptic", "brain", "Doctor"

CANDIDATE REPLACEMENT LIST FOR ENTITIES EXAMPLE:

"regular medication": ["ongoing treatment", "scheduled prescriptions", "routine drugs", "standard therapy", "daily medicine"],
"free": ["complimentary", "at no cost", "without charge", "no fee", "gratis"],
"fit": ["healthy", "suitable", "well-suited", "in shape", "appropriate"],
"life": ["existence", "lifetime", "living", "duration", "human experience"],
"prescribed": ["ordered", "recommended", "allocated", "designated", "authorized"],
"17.0 years": ["17 years", "over 16 years", "17 calendar years", "approximately 17 years", "about 17 years"],
"Anti-epileptic": ["seizure medication", "anti-seizure drugs", "epilepsy treatment", "neurological medication", "anti-convulsant"],
"brain": ["cerebrum", "cranium", "neuroanatomy", "head", "mind"],
"Doctor": ["Physician", "Medical Practitioner", "Healthcare Provider", "Clinician", "Specialist"],

Figure 3: The text that has undergone numerical perturbation, referred to as *Numerical perturbation*, provided by section 3.1, was processed through entity recognition to obtain an entity list. Based on this list, LLMs generated candidate replacement entities. The Candidate replacement list for entities and the text were then selected using embedding model and algorithms to obtain the final protected result.

privacy protection.

4.1. Evaluation Settings

RAG Component: RAG primarily consists of retrieval and generation parts. For the retrieval part, due to its superior performance and ease of use, we adopt the **all-MiniLM-L6-v2** model provided by the sentence-transformers library as the embedding model. This model maintains good performance while reducing computation time and memory usage. We use the Chroma database for storing embeddings and performing similarity queries, leveraging its ease of use and efficient indexing. By default, L2 norm is used to measure vector similarity. For the generation part, we chose widely-used, representative models to ensure generalizable results without being overly powerful, which could overshadow the RAG system’s generative outcomes, making it difficult to evaluate the framework’s effectiveness and privacy protection. Additionally, the selected model should have undergone safety alignment to highlight the vulnerability of RAG systems regarding privacy and the efficacy of our solution.

In this study, we adopt the novel SAGE framework as the primary baseline for comparison. To ensure fairness, the experiments utilize the same datasets mentioned in the original SAGE paper:

- **HEALTHCAREMAGIC:** A dataset comprising 200,000 doctor-patient medical dialogues, reflecting specialized knowledge exchange in the healthcare domain. However, we acknowledge potential biases in this dataset, as it exclusively captures doctor-patient interactions and omits other forms of medical data.
- **WIKITEXT-103:** A 100-million-token Wikipedia-derived corpus, representing general open-domain knowledge scenarios. While this dataset is broadly applicable, we note its limitations in structured information and potential inclusion of outdated or inaccurate content. Since WIKITEXT-103 inherently lacks sensitive data, we follow the methodology of Huang et al. (2023) by injecting privacy information into the public dataset to create a hybrid sensitive corpus. Drawing from SAGE’s data construction approach, we additionally incorporate the *enron-mail* dataset to extract personally identifiable information (PII).

The hybrid dataset is built as follows:

1. Extract PII from *enron-mail* to form a privacy data list.
2. Split WIKITEXT-103 into chunks of 1,500 tokens, then segment each chunk into paragraphs using sentence boundaries (e.g., periods).
3. Insert randomly selected privacy entries from the PII list between paragraphs to create a public-private mixed dataset.
4. Unlike SAGE, we exclude URLs as privacy elements due to their non-sensitive nature and length constraints, substituting them with names.

After deduplication, 42,540 unique privacy entries (from an initial pool of 633,432) are randomly inserted into WIKITEXT-103 segments.

To mitigate dataset-specific randomness, data splits are randomized. However, repeated experiments were omitted due to time and resource limitations. Future work will expand to additional datasets and multiple experimental trials to further validate ENTITYSAFE’s generalization capabilities.

Baseline Methods: Three methods serve as our baselines: the first is generating results directly without RAG, termed 0-shot; the second uses raw data for retrieval, called Origin; and the third adopts the SAGE method, which outperforms simple paraphrasing, ZeroGenYe et al. (2022), and AttrPromptYu et al. (2024).

4.2. Effectiveness Experiments

To evaluate the utility of synthetic data generated by the synthetic data framework as the retrieval component of RAG for the final output of generated text, we selected three metrics for quantitative assessment. The quality of the generated text is measured by comparing its performance against the reference text based on these metrics.

- **ROUGE-L Score:** This metric is used to assess the similarity between the generated text and the reference text, primarily measuring the similarity in word order and content at the sentence level.
- **BLEU-1 Score:** This score calculates the matching rate between the generated text and the reference text at the 1-gram level.
- **Text Similarity:** This metric evaluates the semantic similarity between the generated text and the reference text. The cosine similarity between the generated text and the reference text is calculated as the text similarity.

4.2.1. Utility Results in Medical Dialogues

For the HealthcareMagic dataset, we followed the settings from SAGE Zeng et al. (2024c), prioritizing the splitting of the dataset, where 99% was used as training retrieval data and 1% as validation data. During the evaluation, the user query portion of the validation set was treated as questions, and the doctor’s responses as reference answers. The questions were used to generate text in four different scenarios: the no RAG scenario, the scenario where retrieved data was directly used as the augmentation, the scenario where retrieved content was augmented after SAGE protection, and the scenario where retrieved content was augmented after EntitySafe protection. The metrics were then calculated for each of these four scenarios.

In conducting the experiments, we adopted the experimental setup from the SAGE method. Our approach utilized two tokenization models for entity recognition. The primary model used was SpaCy’s **en-core-web-trf** model, which employs a Transformer-based architecture, enabling it to capture longer dependencies and perform better in understanding context, thereby improving entity recognition. The secondary model

selected was SpaCy’s **en-core-sci-scibert**, which is based on the SciBERT model specifically designed for processing scientific and technical literature. SciBERT is a pre-trained language model for scientific texts, fine-tuned on top of the standard BERT to better capture the language patterns in scientific literature, effectively supplementing the primary model’s shortcomings in the medical domain by identifying entities that the primary model may have missed.

Additionally, given the high privacy requirements in medical scenarios, we set the hyperparameter replacement rate to 50 (i.e., each entity recognized by the secondary model and all nouns in the text identified by SpaCy have a 50% chance of being replaced). Following these settings, we randomly extracted 1,000 dialogue samples from the validation set conduct this part of the experiment. The results are reported in Table 1. It is important to note that the focus of this study is to demonstrate that EntitySafe can achieve comparable performance to SAGE while mitigating privacy leakage risks, showcasing its feasibility in practical applications. Therefore, the four significant digits in Table 1 are intended to more clearly illustrate the subtle differences between EntitySafe and SAGE, allowing for a more accurate evaluation of their performance.

Table 1: Comparison of the Effectiveness of ours with SAGE on the HealthcareMagic dataset

Method	BLEU-1	ROUGE-L	Similarity
0-shot	0.1322	0.0981	0.5404
Origin	0.1539	0.1137	0.5709
SAGE	0.1109	0.1495	0.5693
ours	0.1114	0.1497	0.5650

The results indicate that after addressing the potential privacy leakage risks associated with directly transmitting private data to a closed-source large model in the SAGE method, our synthetic data solution maintains performance comparable to that of the SAGE approach across three metrics. In fact, we even achieve slightly higher scores in BLEU-1 and ROUGE-L, while the difference in similarity metrics is within 1%, significantly outperforming methods without RAG.

This demonstrates that using an RAG system can effectively enhance the quality of model responses in vertical domains. Furthermore, in this context, our method exhibits a high degree of consistency with the SAGE method, showing slightly superior performance in two of the metrics. This may be attributed to the fact that SAGE generates text anew, while we employ a probabilistic replacement approach, leading to a slight advantage in text similarity but a minor disadvantage in semantic similarity. This suggests that our synthetic data generation solution is on par with SAGE in terms of preserving the utility of the original data.

The slight superiority in text similarity and minor deficiency in semantic similarity may be attributed to the fact that SAGE employs text regeneration, whereas our method uses a probabilistic replacement strategy. This suggests that our synthetic data generation approach matches SAGE in preserving the utility of original data while showing nuanced differences in how

text similarities are handled.

”While our effectiveness experiments demonstrate that EntitySafe generally preserves the utility of RAG, we acknowledge that the performance may vary depending on the type of query. Although our experimental design did not explicitly categorize queries based on their numerical precision requirements, a post-hoc analysis suggests that queries involving precise numerical values (e.g., ”What is the exact dosage of this medication?”) may experience a slight reduction in accuracy compared to queries involving ranges or trends (e.g., ”What is the typical range for blood pressure in patients with this condition?”). This is because the noise introduced by our perturbation mechanism can inevitably affect the accuracy of precise numerical values. However, we believe that this trade-off is acceptable in many real-world scenarios, where a certain level of imprecision is tolerable in exchange for stronger privacy protection. Moreover, for applications requiring extremely high numerical accuracy, alternative privacy-preserving techniques or a re-evaluation of the need for RAG may be necessary.”

4.2.2. Utility Results in General Dialogue

To evaluate performance in general dialogue, we used the generated wiki_pii dataset as the retrieval data for RAG, as described in the experimental setup in Section 4.1. Given the vastness of human knowledge, we aimed to avoid posing questions that are entirely unrelated to the experimental dataset, which could render RAG ineffective in this context. In such cases, the model might disregard the retrieved content and generate results independently. For example, if a question is asked that is not covered in the dataset, the model might respond as follows:

Example

The information you provided does not contain any information about the New Orleans Saints or their Super Bowl wins.
However, I can provide you with the answer....

Subsequently, the model will respond based on its existing memory. While the final responses may yield good results according to the metrics, they do not validate our objective of assessing the effectiveness of retrieval data under synthetic data protection for the overall RAG results. Therefore, if the testing domain of questions is significantly broader than the knowledge domain contained in the wiki_pii dataset, the experimental results may lose their value. To ensure that the knowledge domain contained in the wiki_pii dataset adequately covers the testing question domain, we generated a new QA dataset using a large model based on the existing wiki_pii dataset.

Specifically, we randomly selected 250 segments from the segmented text and utilized the gpt-4o-mini model to generate questions based on the provided wiki segments, along with their corresponding answers. This resulted in the creation of a wiki_QA dataset, on which we subsequently conducted effectiveness experiments. As before, we adopted the settings from the SAGE method as described in their paper. In using our approach, we considered the current scenario and determined that

privacy requirements were relatively low for this task, as the general dialogue contained less specialized knowledge. Consequently, we set the hyperparameter replacement rate to 15.

To simultaneously assess the impact of the hyperparameter "replacement rate" on effectiveness, we conducted comparative experiments with varying replacement rates while keeping other conditions constant. We selected 50 as the replacement rate for the second group of experiments. The experiment with a replacement rate of 15 is referred to as ours_1, while the one with a replacement rate of 50 is referred to as ours_2. The experimental results are presented in Table 2.

Table 2: Comparison of the Effectiveness of ours with SAGE on the wiki pii dataset

Method	BLEU-1	ROUGE-L	Similarity
0-shot	0.2390	0.2486	0.6442
Origin	0.3432	0.3671	0.7155
SAGE	0.2135	0.2145	0.6473
ours_1	0.2757	0.2943	0.6882
ours_2	0.2556	0.2645	0.6821

Similar to Table 1, the use of our proposed synthetic data as retrieval data demonstrates consistently high performance whether the replacement rate is 15 or 50, albeit slightly inferior to directly using the original data for retrieval. The SAGE framework, however, performs poorly under these experimental conditions, even underperforming compared to not using RAG, possibly due to convergence difficulties in the generation and rewriting stages caused by the insertion of substantial privacy content. Although it achieves relatively higher semantic performance by extracting key information, it still significantly lags behind our approach.

Our method employs partial entity replacement, preserving a high degree of the original text structure and specific tokens while replacing critical information, leading to commendable performance in F1 scores and BLEU scores. Furthermore, by selecting candidate replacement entities with BERT embeddings that have a close similarity to the original text during the final replacement entity determination, we achieve satisfactory performance in semantic similarity metrics. This ensures that LLMs can still produce results of relatively good quality when using our protected data as retrieval data.

As the replacement rate increases, all three types of metrics show a decline compared to lower replacement rates, which aligns with commonsense reasoning; increasing the amount of replaced text naturally reduces similarity with the original text. However, from the numerical changes observed, while BLEU-1 and ROUGE-L scores drop significantly, the change in semantic similarity is minimal, indicating that even with increased replacement rates, a considerable degree of semantic similarity with the original text is maintained. In other words, our method maintains high effectiveness even under scenarios with high replacement rates.

Our method employs partial entity replacement, meticulously preserving the original text structure and specific tokens while substituting critical information. This strategy yielded

commendable scores in both ROUGE-L and BLEU metrics. Furthermore, by selecting candidate replacement entities with BERT embeddings that closely resemble the original text’s similarity, our approach also achieved satisfactory results in semantic similarity. Consequently, this allowed the LLM to generate quality outputs even when using our privacy-protected data as retrieval input.

In summary, the utility results from medical dialogues demonstrate that employing our protection strategy-generated synthetic data as retrieval augmentation significantly enhances model output quality in specialized vertical domains, outperforming non-RAG scenarios and nearly matching the SAGE method’s performance. Experiments in general dialogue further evidenced the ability of both protective strategies to retain original text information when RAG retrieval precisely matched the target. In this context, the SAGE method’s text reorganization through rewriting led to lower ROUGE-L and BLEU scores but slightly higher semantic similarity compared to non-RAG scenarios. Conversely, our method consistently outperformed both non-RAG setups and the SAGE approach across all three metrics, indicating a superior retention of original text and semantic information, thereby providing more effective RAG support.

4.3. Privacy Experiments

To evaluate the privacy protection capability of synthetic data generated by the synthetic data framework as part of the RAG retrieval component for the final output of generated text, Nicholas Carlini et al (Carlini et al. (2023)). found that even large language models, which are trained on vast amounts of data and are less prone to overfitting, can "remember" and leak portions of their training data. They also proposed a simple yet effective attack method that can extract text sequences from the training data merely by querying the language model. Based on this, Zeng et al. (Zeng et al. (2024b)) proposed directed and untargeted Attack to extract private content from the RAG system’s retrieval database. We referenced the methods of Zeng et al. (Zeng et al. (2024b)) to conduct directed and untargeted attacks. The attack prompts they proposed consist of two components: information and command. The information component specifies the target information for the attack, while the command component instructs the model on how to respond after acquiring the target information.

Untargeted attacks refer to scenarios where the attacker does not aim to extract specific training data samples but instead tries to extract as many training data samples as possible from the LLM. The attacker is not concerned with the content of the extracted samples but rather with the LLM’s memory capability. In contrast, targeted attacks focus on specific sample content of interest to the attacker.

4.3.1. Untargeted Attacks

we compared the final output results of the generative model with data protection measures against the original data without synthetic data protection. It was determined that if there are at least 10 tokens that match exactly in both pieces of text, then the non-directed attack has indeed extracted information

from the original text. We then counted the number of successful non-directed attack prompts that achieved information extraction, which we refer to as "repeat prompts." According to SAGEZeng et al. (2024c), they also calculated the ROUGE-L score, considering that if this score exceeds 0.5, it indicates a high similarity between the generated results and the original text, confirming that information from the original text has indeed been extracted. The privacy experiments were conducted using the same experimental methods as those used in the effectiveness experiments. The experimental results are shown in Table 3.

Table 3: Comparison of ours with SAGE under untargeted attack on the wiki_pii dataset

Method	repeat prompt	repeat context
Origin	173	23
SAGE	90	0
ours_1	172	15
ours_2	78	1

From the experimental results, it can be observed that the approach using synthetic data protection methods can significantly reduce potential privacy leakage issues. While ours_1 underperforms compared to the SAGE method on both test metrics, the ours_2 method, which has an increased substitution rate, outperforms the SAGE method on the repeat prompt metric and shows comparable performance on the repeat context metric. This indicates that increasing the substitution rate can effectively enhance the method’s capability to protect privacy, providing more robust privacy protection than the sage method in scenarios with a high substitution rate.

4.3.2. targeted Attacks

The test metrics selected for targeted attacks were data extraction and repeat prompt, where data extraction refers to the number of privacy information attacks successfully extracted from the model via targeted attacks. Specifically, we inserted content extracted from enron-mail into the wiki dataset to construct the wiki_pii dataset, serving as a candidate privacy repository. Through experimentation, we evaluated how much of the extracted content via targeted attacks could fully match within the privacy repository. An attack was considered successful if it could extract at least three segments of completely matching information. The evaluation rule for repeat prompt remained consistent with that in untargeted attacks. Experimental results are shown in Table 4.

Table 4: Comparison of ours with SAGE under targeted attack on the wiki_pii dataset

Method	target info	repeat prompt
Origin	51	180
SAGE	0	47
ours_1	13	176
ours_2	2	30

The results of the targeted attacks draw similar conclusions to those of the untargeted attacks. Comparative experiments reveal that increasing the substitution rate significantly reduces the number of successful attacks extracting private information. This verifies the capability of our method in enhancing privacy protection.

Combining the aforementioned experiments proves that adjusting hyperparameters and selecting entity recognition models can allow users to dynamically balance privacy and effectiveness according to application scenarios. This makes our method applicable in a wider range of different contexts.

4.4. Ablation Experiments

To investigate the impact of various components that make up the entire synthetic data framework on the final results, we conducted ablation experiments in this section. Specifically, we examined the effects of altering or omitting two key components: the Value perturbation Component and the Entity Recognition Component. To investigate the role of these components within the entire data synthesis framework.

4.4.1. Value perturbing Component

To assess the influence of the data perturbing component on the experimental outcomes, we omitted this component and directly replaced entities in the original retrieval text. We employed the same settings described in Sections 4.2 and 4.3 to evaluate the performance differences on effectiveness and privacy metrics after excluding the value perturbation component.

The experimental results regarding the effectiveness are shown in Figure 4, and the experimental results regarding privacy are illustrated in Figure 5.

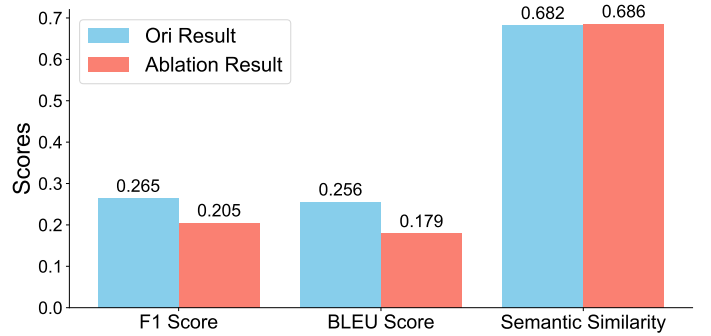


Figure 4: Ablation Study on the Effectiveness of the Numerical perturbation Module

The results reveal an unexpected decline in the F1 and BLEU scores after the removal of the component. However, a combined analysis with the enhanced similarity scores indicates that since the F1 and BLEU metrics emphasize exact text matching, the rephrasing of original numerical elements during entity substitution led to a lack of precise correspondence with the reference texts. Despite this rephrasing, the semantic meaning is not significantly affected, resulting in an anticipated increase in similarity compared to cases without numerical perturbation. The experiment demonstrates that the numerical perturbation

module can achieve a notably good preservation of semantic information; even when numerical perturbation is applied, it does not have a substantial negative impact on validity.

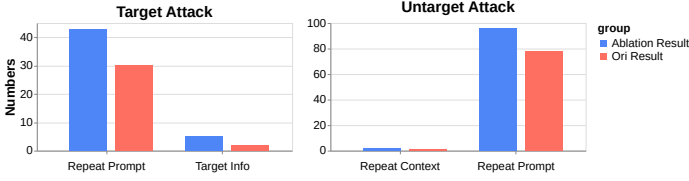


Figure 5: Ablation Study on the Privacy of the Numerical Module

The results of the privacy experiments demonstrate that numerical perturbation exhibits protective effects on privacy in both the target attack and untargeted attack experiments. Across the four metrics evaluated in both experimental setups, the inclusion of the numerical perturbation module has been shown to enhance privacy preservation.

In summary, the numerical perturbation module within the overall data synthesis framework can significantly reduce the likelihood of privacy leakage with minimal impact on semantic integrity.

4.4.2. Entity Recognition Component

To assess the influence of the entity recognition module on experimental results, a three-part experiment was conducted: The auxiliary model and noun substitution components were removed from the dual-model entity recognition framework, retaining only the primary model’s recognition outcomes. we called it **situation_1**. Entity recognition by the auxiliary model was disabled while keeping the primary model’s recognition results and the noun identification outcomes. we called it **situation_2**. Both the primary and auxiliary models were deactivated, preserving only the results of noun substitution. we called it **situation_3**.

These experiments were designed to isolate the contributions of each component within the entity recognition system, aiming to understand their individual impacts on the overall performance and accuracy of entity recognition.

During the aforementioned experiments, all other settings remained consistent with those used in the evaluation of the numerical perturbation component. The experimental results concerning validity are illustrated in Figure 6, while the outcomes related to privacy are presented in Figure 7.

The experimental results show that as the extent of substitution gradually decreases, there is an upward trend in the F1 score, BLEU score, and semantic similarity. Concurrently, the risk of privacy leakage increases rapidly. This trade-off is particularly evident when only nouns are substituted. The experimental outcomes indicate that each component within the entity recognition process plays a part in constructing the entire list of substitute entities. The collaborative effect of multiple components achieves a well-balanced compromise between privacy and validity, performing admirably across various metrics. Additionally, through this segment of experiments, it has been observed that besides modulating the substitution rate, the balance

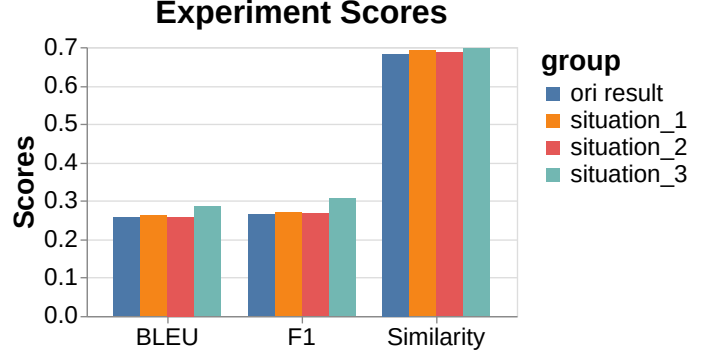


Figure 6: Ablation Study on the Effectiveness of the Entity Recognition Component

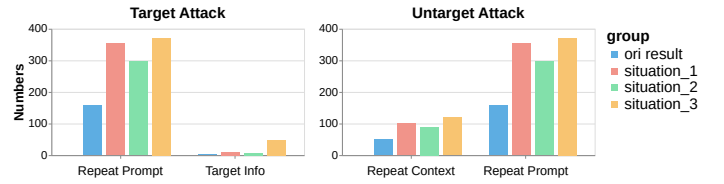


Figure 7: Ablation Study on the Privacy of the Entity Recognition Component

between validity and privacy can also be adjusted by varying the number of models used and whether to substitute nouns, thereby catering to specific requirements.

5. Discussion

In this work, we have introduced *EntitySafe*, the first framework to combine differential privacy-driven numerical perturbation with probabilistic entity replacement for safeguarding sensitive information in Retrieval-Augmented Generation (RAG) systems. Through extensive experiments on both HealthcareMagic (medical dialogues) and WikiPII (general-domain with injected PII) datasets, we demonstrated that *EntitySafe* preserves retrieval utility—achieving BLEU-1 and ROUGE-L scores within 1–2% of the SAGE baseline—while reducing successful extraction prompts in targeted and untargeted attack scenarios by over 80%. These results confirm that flexible, parameterizable trade-offs between data utility and privacy protection can be achieved in practical RAG deployments.

5.1. Future Work & Outlook

- **Formal Security Analysis.** While our numerical perturbation module provides rigorous (ϵ, δ) -differential privacy guarantees for continuous attributes, the probabilistic entity replacement mechanism currently lacks an analogous formal treatment. We plan to develop a comprehensive theoretical framework—potentially based on Rényi differential privacy (Mironov, 2017) or f -divergence metrics—to quantify privacy leakage, establish end-to-end privacy budgets, and guide principled parameter selection.

- **Comprehensive Adversarial Benchmarking & Model Independence.** Our privacy evaluation to date has focused on single-shot membership and extraction attacks. Future assessments will incorporate multi-round probing, attribute inference, and collusion scenarios to better reflect real-world threat models. Concurrently, we will investigate locally deployable or distilled language models and rule-based substitution methods to reduce reliance on external LLM APIs, thereby minimizing trust assumptions and potential side channels.
- **System and Hardware Optimization.** The current two-stage *EntitySafe* pipeline incurs an average latency of tens of milliseconds per document on a single GPU. To meet ultra-low-latency or high-throughput requirements, we will explore batching, cache reuse, noise precomputation, and hardware acceleration (e.g., GPU/TPU kernel optimization) to minimize end-to-end response times and resource consumption.
- **Domain Adaptation & Human-Centric Evaluation.** To validate generalizability, we will extend our experiments to additional verticals such as finance, legal, and government datasets, and conduct comparative studies against emerging privacy-preserving techniques. Moreover, user studies will assess the perceptual quality and acceptability of protected texts, informing guidelines for setting privacy-utility parameters in real-world applications.

By pursuing these research directions, we aim to evolve *EntitySafe* into a robust, theoretically grounded, and practically viable privacy enhancement toolkit for RAG systems, enabling secure and trustworthy AI applications across diverse domains.

6. Conclusion

In this paper, we explore the effectiveness of using entity replacement as a data synthesis scheme for protecting privacy data in the RAG context. We propose a data synthesis approach that allows for the adjustment of the replacement rate and the selection of entity recognition models to achieve a balance between utility and privacy requirements for different privacy scenarios. By employing numerical fuzziness and entity perturbation, we effectively disrupt the attacker’s inference chain to protect privacy data. Moreover, since the selection of replacements is determined by semantic similarity, we maintain a high level of usability while safeguarding private content. Our work provides new methods and insights for privacy protection in RAG systems.

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